

AI-Powered Arabic Language Tutor for Autistic Children (Noor)

Graduation Defense Presentation

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Introduction & Motivation



Fusion 360 Prototype Design

What is Noor?

An AI-powered Arabic tutor for children with Autism Spectrum Disorder in Arabic-speaking communities.

Multi-Modal Design

Combines voice interaction, visual perception, animated face interface, and physical head movement.

Cultural Fit

Addresses Arabic dialect and cultural needs often ignored by Western-designed tools.

Accessible Prototype

Built with affordable embedded hardware for real-world interaction in families and centers.

Problem Statement

Limited Arabic Tools

Few affordable, interactive language-learning tools exist for Arabic-speaking autistic children.

High Cost

Imported therapy robots like NAO and Pepper are prohibitively expensive for most families and centers.

Passive Screens

Screen-only apps lack physical engagement, eye-contact simulation, and natural turn-taking.

Cultural Gap

Most systems use English or Modern Standard Arabic, not the spoken dialects children use at home.

Project Aim

To design and implement an AI-powered interactive Arabic tutor that supports language practice and social engagement for children with Autism Spectrum Disorder.

1

Interactive Prototype

Physical robot with embedded hardware for real-world interaction.

2

Arabic AI Pipeline

End-to-end voice pipeline: STT
→ LLM → TTS using cloud APIs.

3

Adaptive Behavior

Emotion-aware responses and reinforcement learning for personalized engagement.

Project Objectives



Interactive Prototype

Build a physical Noor prototype using Jetson, ZED 2, ReSpeaker, Arduino, and servo motors.



Vision & Emotion

Implement face tracking and emotion recognition for adaptive, child-aware interaction.



Arabic Voice Pipeline

Develop a full STT → Gemini → TTS pipeline using Groq Whisper and ElevenLabs APIs.



Safe Movement

Provide child-friendly responses with smooth, limited-angle servo motion and smart interruption.

Noor: A Multi-Modal AI Companion



Arabic Speech Interaction

ReSpeaker captures voice input, Groq Whisper transcribes it, and ElevenLabs generates Arabic speech output.



Visual Perception

ZED camera supports face tracking, child localization, and emotion-aware interaction.



Animated Face Interface

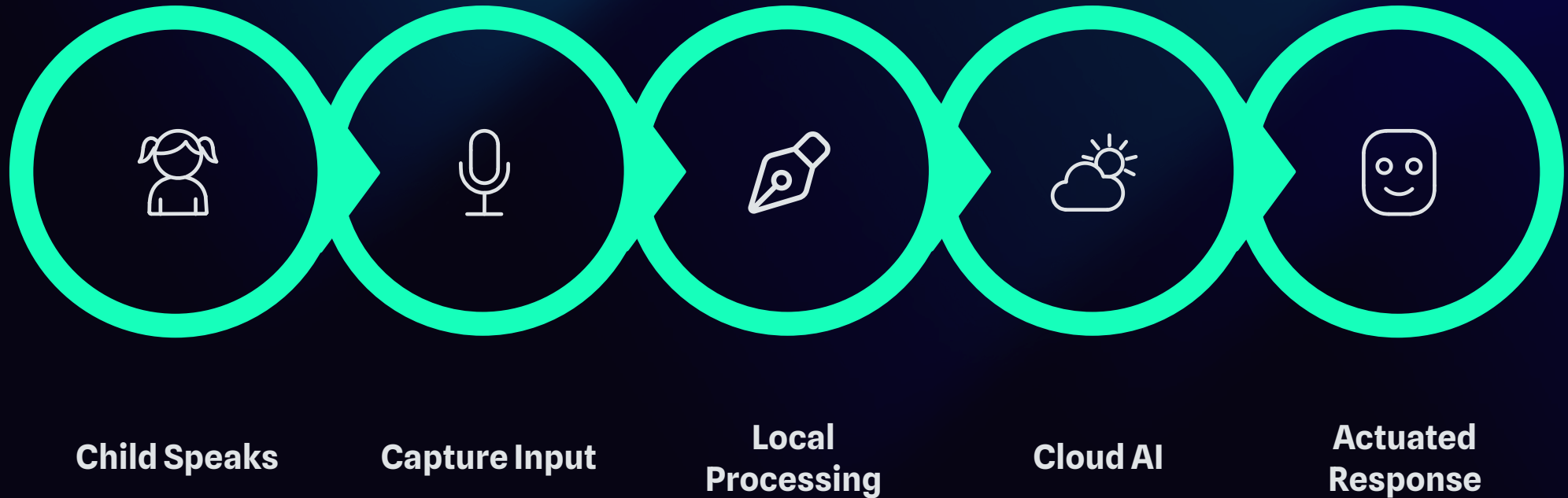
The display shows Noor's idle, listening, and speaking states.



Servo Head Movement

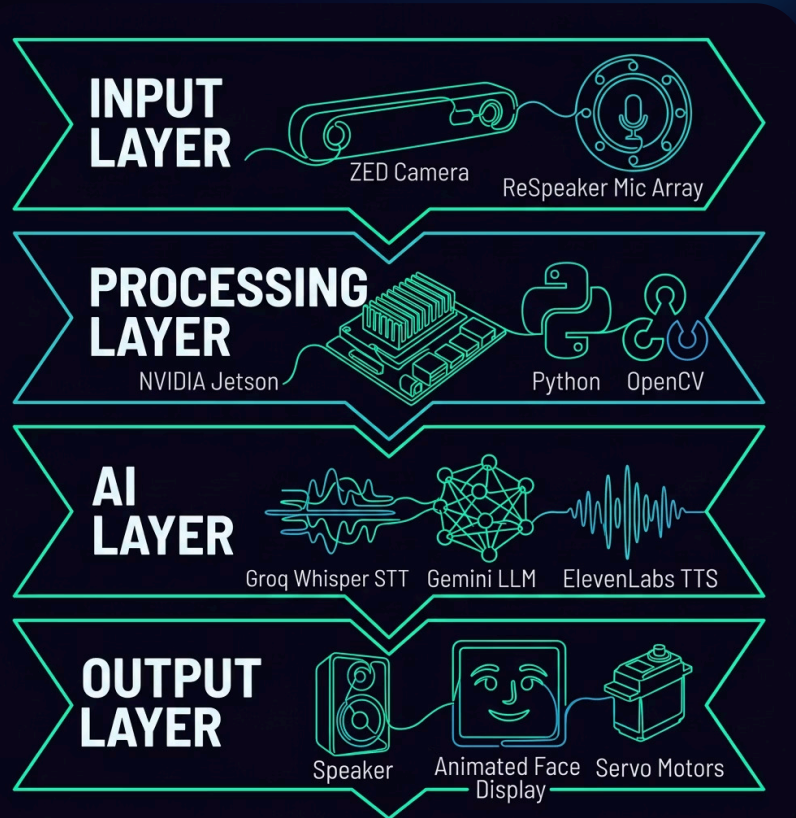
Arduino-controlled servo motors orient Noor toward the child using face or sound direction.

System Concept Flow



Voice & face input → Cloud AI processing → Coordinated response output

High-Level System Overview



Layer Breakdown

- **Input:** ZED 2 camera and ReSpeaker USB Mic Array V3.1 capture audio and visual signals.
- **Processing:** NVIDIA Jetson runs Python modules for VAD, tracking, and coordination.
- **AI:** Cloud APIs handle Arabic STT, language reasoning, and TTS synthesis.
- **Output:** Speaker, animated face display, and servo motors deliver the response.

Hybrid Edge-Cloud Architecture

Edge Side — Local

- NVIDIA Jetson + Python core
- ZED 2 Camera + OpenCV tracking
- ReSpeaker Mic + WebRTC VAD
- Arduino Uno + Servo Motors
- Animated Face UI (display)

Cloud Side — APIs

- Groq Whisper — Arabic STT
- Gemini API — Language reasoning
- ElevenLabs API — Arabic TTS

 Hybrid design balances real-time edge responsiveness with high-quality Arabic AI in the cloud.]